

Biochemistry Lec.1

- Classification
- Monosaccharides



General Biochemistry

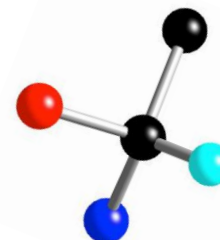
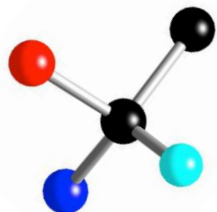
1- Carbohydrate chemistry

2- Protein chemistry

3- Nucleic acid chemistry

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Head of Biochemistry Department

CHEMISTRY OF CARBOHYDRATES



Objectives

To understand:

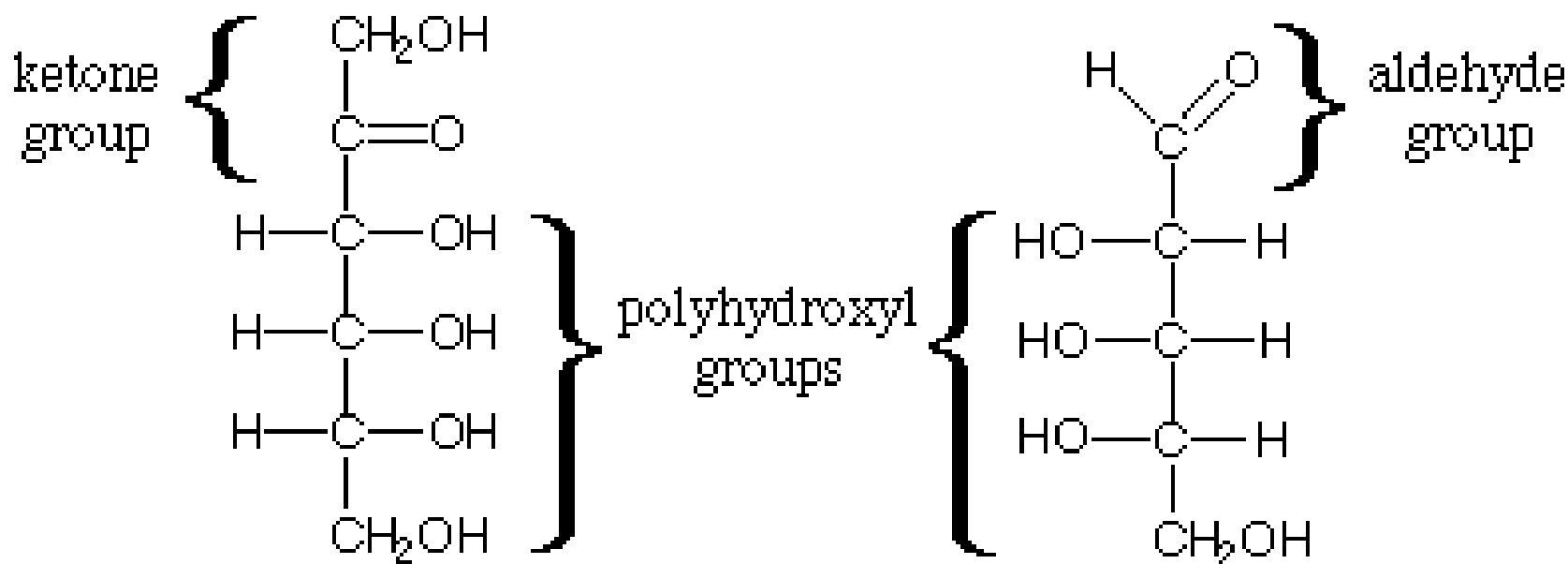
- 1- Nature and definition of carbohydrates
- 2- The basic structure and properties of carbohydrates
- 3- Classification of carbohydrates
- 4- Functions of important carbohydrates

When you hear the word ‘carbohydrate’ what comes to your mind?

- ✓ Carbohydrates are “**Hydrates of Carbon**” $\text{C H}_2\text{O}$
- ✓ Carbohydrates are organic biomolecules abundantly present in the nature.
- ✓ Found in the cells of plants and animals.
- ✓ Structural formula/General formula for simple carbohydrates : $\text{C}_n (\text{H}_2\text{O})_n$
Where n = **number of carbon atom** present in carbohydrate structure.
- ✓ **Old definition is not valid since:**
 - Certain Carbohydrates:**
Rhamnose did not fit in the structural formula of carbohydrates.
 - Certain non Carbohydrates:**
Lactate and Acetate fitted in the structural formulae.

New Definition of Carbohydrates

- ✓ Chemically composed of **Polyhydroxy Aldehyde or Polyhydroxy Ketone** or their condensed products or their derivatives.
- ✓ Some carbohydrates also contain nitrogen, phosphorus or sulfur.
- ✓ They are referred as **saccharides** from the latin word *saccharum* which means 'sugar'.
- ✓ The simpler members of the carbohydrates family often give a sweet taste. They have a suffix *-ose*



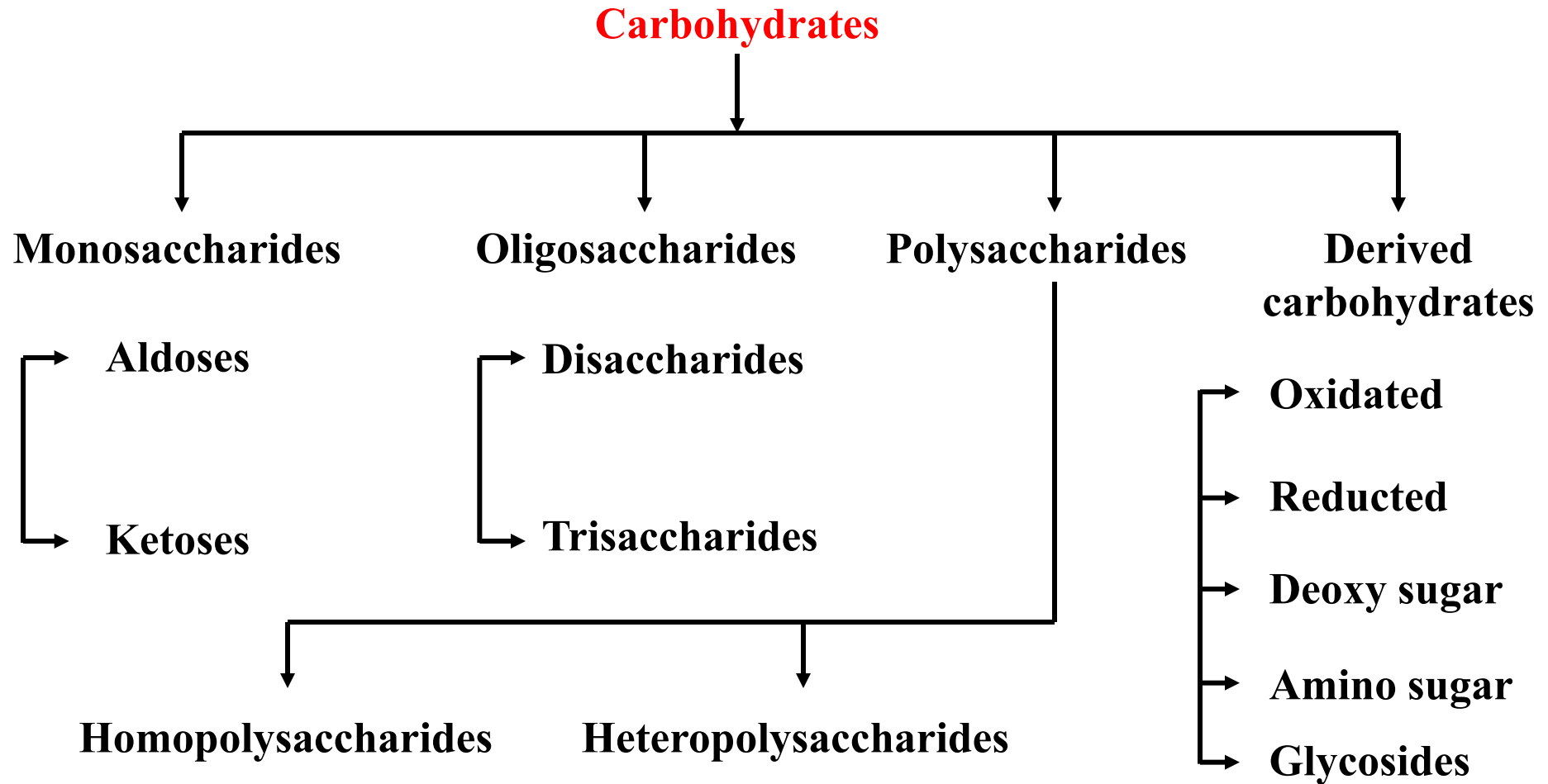
Carbohydrates



- General functions:

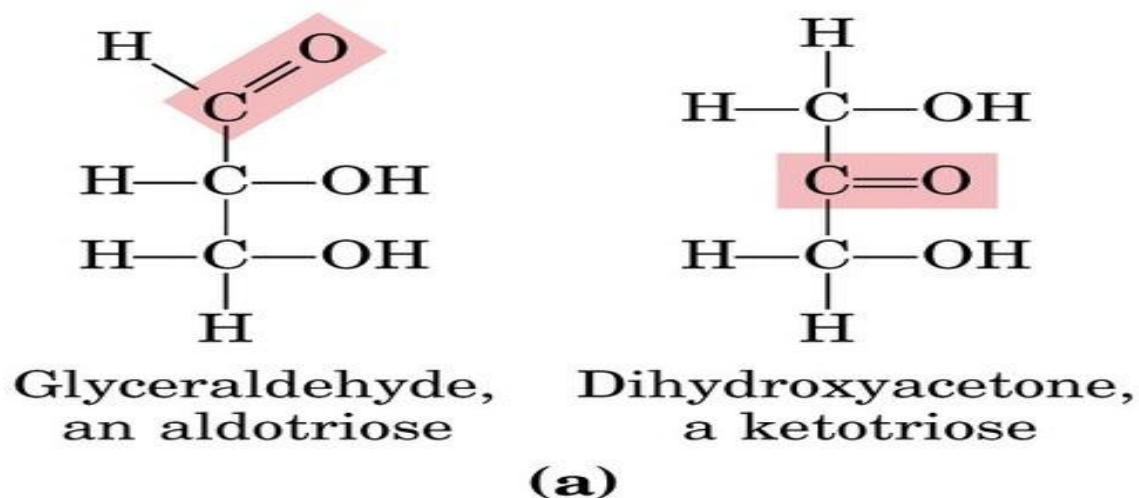
- ✓ The major source of energy for plant, animals and human 4 Kcal/g
- ✓ Essential part of many important molecules (ATP, DNA..)
- ✓ Supportive structural components such as polysaccharides (e.g. cell walls of bacterial and plant)
- ✓ Linked to proteins and lipids (glycoconjugates)
 - In the extra cellular matrix, they exert effects on **cellular recognition** in **infection, cancer, and immune response**.
- ✓ ABO blood markers found on red blood cells are made up of carbohydrates. They allow us to distinguish our body's blood type from a foreign blood type.

Classification of carbohydrates



Monosaccharides

- ✓ Monosaccharides are the simplest form of carbohydrates with at least 3 carbon atoms.
- ✓ They are colorless crystalline solids that are freely soluble in water but insoluble in non polar solvents (alcohol, ether etc.).
- ✓ They are an **unbranched carbon chain** where all the carbon atoms are linked by single bond.
- ✓ They consist of a single **polyhydroxy aldehyde** or **ketone** unit.
- ✓ One of the carbon atom is double bonded to form a functional group named **carbonyl group**; each of the other carbon atom has a hydroxyl group.
- ✓ Depending upon the position of the carbonyl group they are either **aldose** or **ketose**.

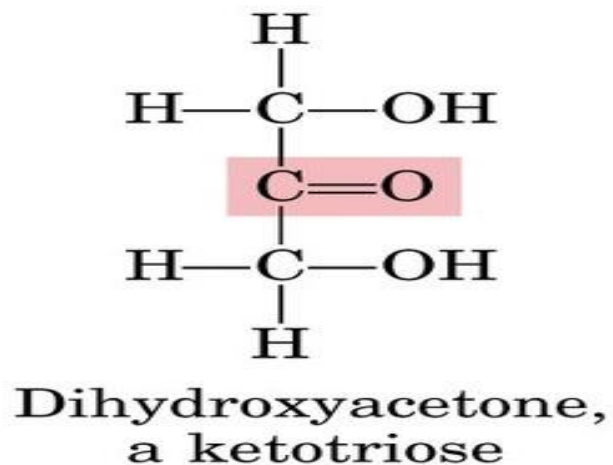
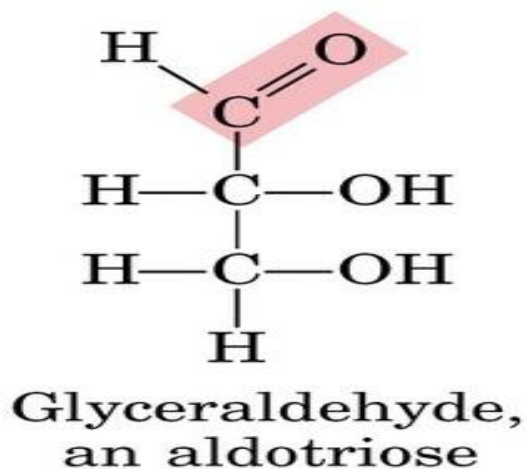


Fischer Projections

Fischer projections allow us to represent two dimensional organic molecular structure.

In Fischer projections:

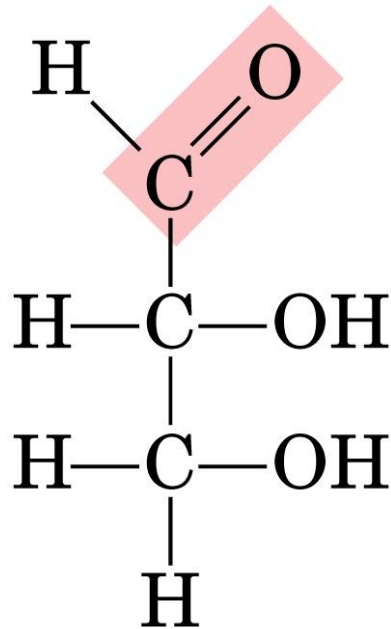
- ✓ Functional group is always at the top
- ✓ Carbon chain is written vertically with substituent groups to the left and right
- ✓ Uses horizontal lines for bonds (hydrogen – hydroxyl group)
- ✓ Uses vertical lines for bonds (carbon – carbon)



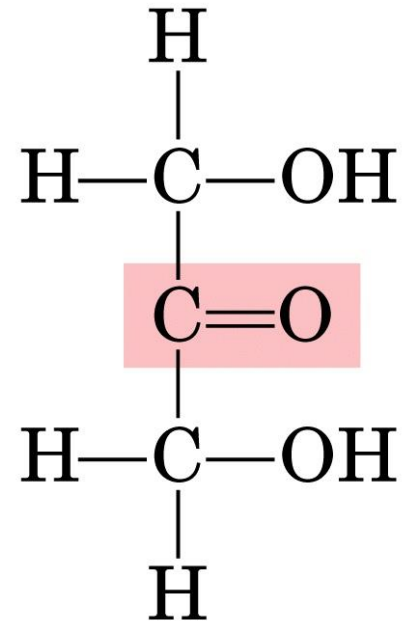
(a)

Number of Carbon Atoms	Aldoses (Aldehyde-CHO)	Ketoses (Ketone -C=O)
3 Triose	Aldo Triose Glyceraldehyde	Keto Triose Di HydroxyAcetone
4 Tetrose	Aldo Tetrose Erythrose	Keto Tetrulose Erythrulose
5 Pentose	Aldo Pentose Ribose, Xylose, Arabinose	Keto Pentulose Ribulose, Xylulose
6 Hexose	Aldo Hexose Glucose, Galactose ,Mannose	Keto Hexose Fructose
7 Heptose	Aldo Heptose SedoHeptose	Keto Heptulose SedoHeptulose

Triose



Glyceraldehyde,
an aldotriose

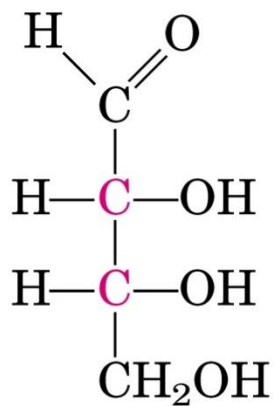


Dihydroxyacetone,
a ketotriose

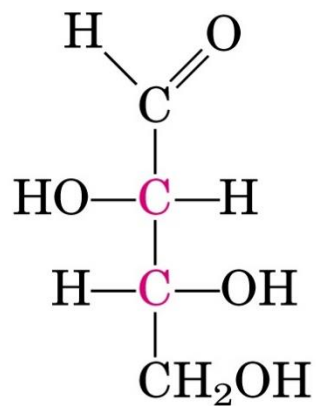
(a)

Tetrose

Four carbons

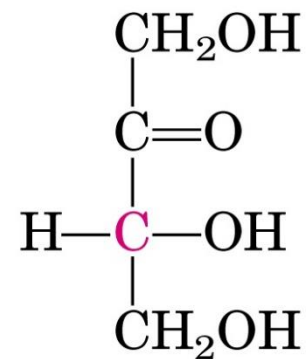


D-Erythrose



D-Threose

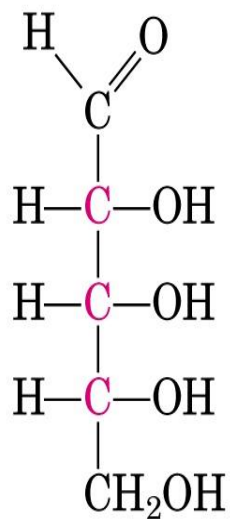
Four carbons



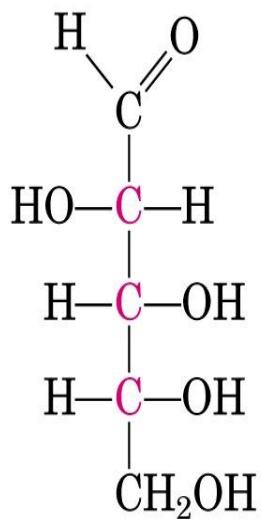
D-Erythrulose

Pentose

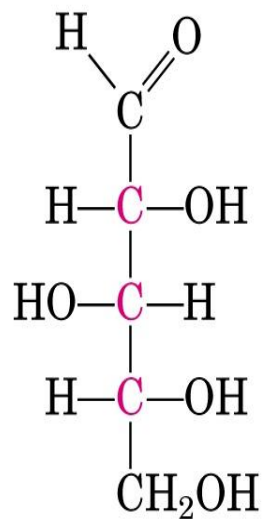
Five carbons



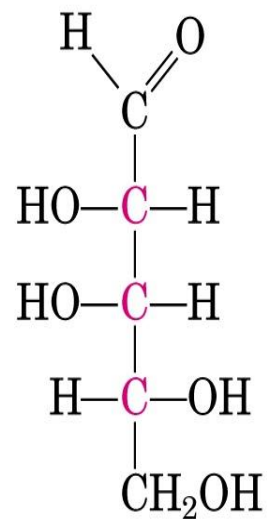
D-Ribose



D-Arabinose

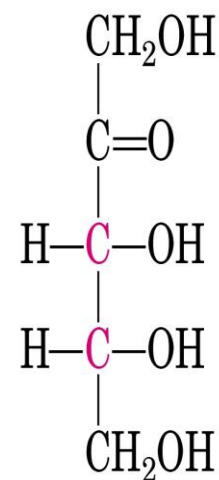


D-Xylose

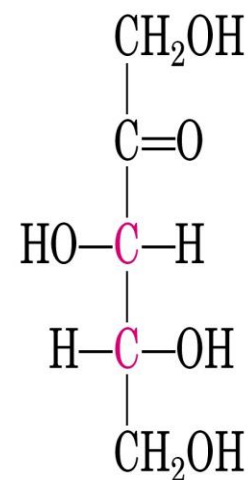


D-Lyxose

Five carbons



D-Ribulose



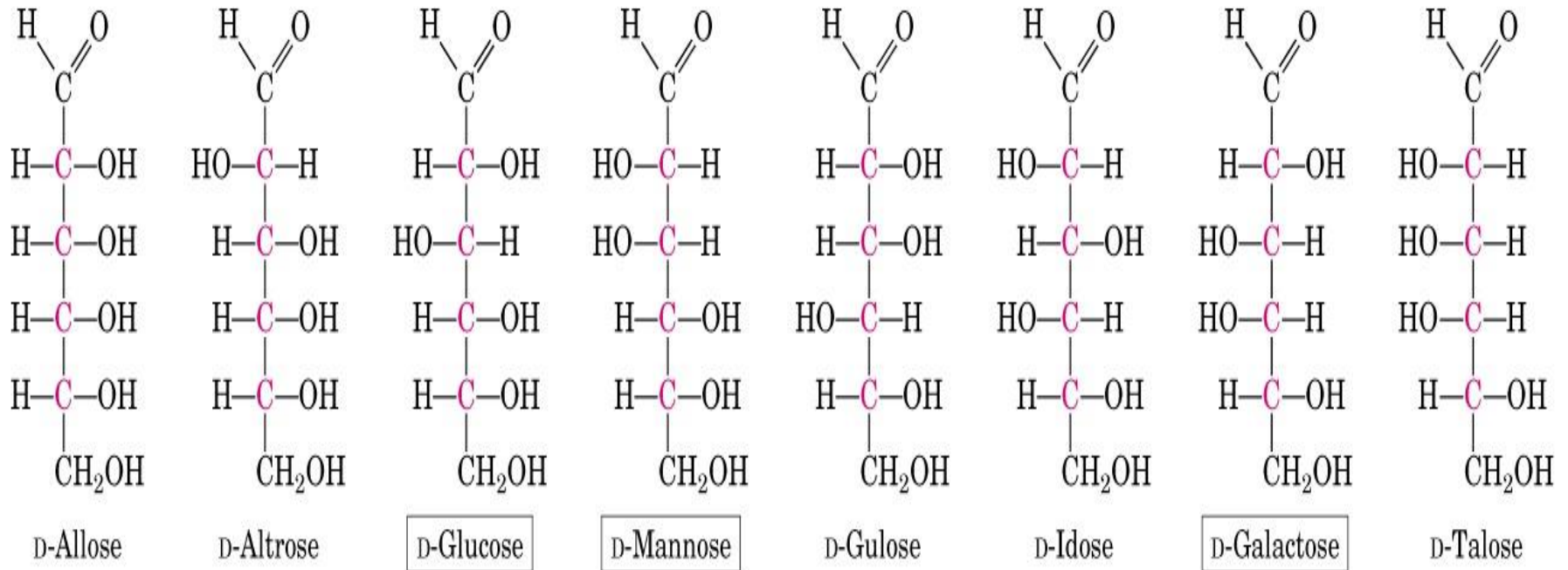
D-Xylulose

Importance of pentose

- ✓ Ribose and deoxyribose enter in the structure of nucleic acids.
- ✓ Ribose enters in the structure of coenzymes NAD, NADP and flavoproteins.
- ✓ Ribose enters in the structure of ATP, GTP and other high energy phosphate compounds.
- ✓ Ribose phosphate and ribulose phosphate are intermediates in pentose phosphate shunt (a minor pathway for glucose oxidation).
- ✓ Arabinose and xylose are constituents of glycoproteins in plants and in animals.
- ✓ Lyxose present in bacterial glycolipids.
- ✓ Xylulose is an intermediate in uronic acid pathway (a minor pathway for glucose oxidation).

Hexose

Six carbons

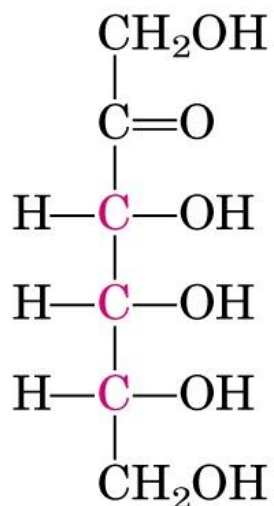


D-Aldoses

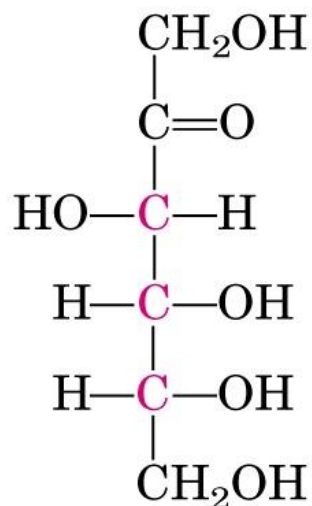
(a)

Hexose

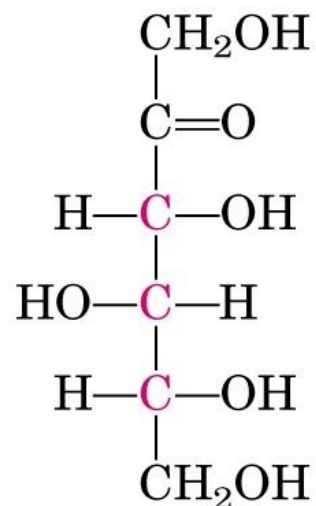
Six carbons



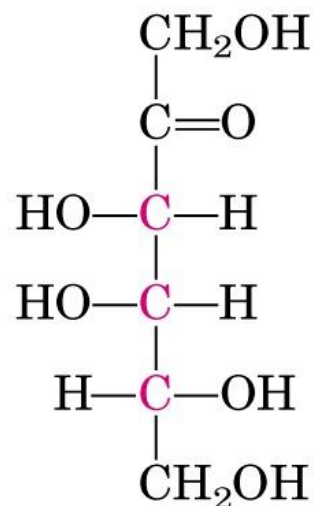
D-Psicose



D-Fructose



D-Sorbose



D-Tagatose

D-Ketoses (b)

Importance of hexose

- ✓ **Glucose (blood sugar or grape sugar) is the most important sugar in carbohydrate. Glucose is the major source of energy in mammals.**
- ✓ **Fructose (fruit sugar) can be converted into glucose in the liver.**
It is the main sugar of semen, also it termed fetal sugar.
- ✓ **Galactose can be converted into glucose in the liver.**
It is synthesized in mammary gland to make the lactose of milk.
- ✓ **Mannose is a constituent of many glycoproteins and glycolipids.**
- ✓ **Idose not found in nature but its acid derivative present in dermatan sulphate and heparan sulphate.**
- ✓ **Talose has antimicrobial and anti-inflammatory activities.**
- ✓ **Allose has antioxidant activity.**
- ✓ **Altrose is a constituent of bacterial cell wall.**
- ✓ **Sorbose used commercially for synthesis of vitamin C, and has sweetness equivalent to sucrose.**
- ✓ **Tagatose reduces blood glucose level (good artificial sweetener).**
- ✓ **Psicose is good for mental health, reduce blood glucose level, has low caloric value**